Tutorial 2

Prove the following statement.

Theorem (Hessian test for convexity). Let $C \subseteq \mathbb{R}^d$ be open and convex, and let $f : C \to \mathbb{R}$ be twice continuously differentiable. Then

$$f \text{ is convex on } C \quad \Longleftrightarrow \quad \nabla^2 f(x) \succeq 0 \qquad \text{for every } x \in C.$$

Solution: ($\Longrightarrow$) Assume that f is convex. Since f is differentiable, the first-order characterization of convexity gives

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \qquad \text{for all } x, y \in C. \tag{1}$$

Write (1) once as it is, and once with x and y interchanged:

$$f(x) \geq f(y) + \nabla f(y)^T (x - y).$$

Adding the two inequalities, the terms $f(x)$ and $f(y)$ cancel:

$$0 \geq \nabla f(x)^T (y - x) + \nabla f(y)^T (x - y) = \big(\nabla f(x) - \nabla f(y)\big)^T (y - x).$$

Hence the gradient is monotone:

$$\big(\nabla f(y) - \nabla f(x)\big)^T (y - x) \geq 0 \qquad \text{for all } x, y \in C. \tag{2}$$

Fix $x \in C$ and $v \in \mathbb{R}^d$. Because C is open, $x + tv \in C$ for all sufficiently small $t > 0$. Put $y = x + tv$ in (2), so $y - x = tv$:

$$\big(\nabla f(x + tv) - \nabla f(x)\big)^T (tv) \geq 0.$$

Dividing by $t^2 > 0$,

$$\frac{\big(\nabla f(x + tv) - \nabla f(x)\big)^T v}{t} \geq 0. \tag{3}$$

Letting $t \downarrow 0$ and using differentiability of ∇f,

$$v^T \nabla^2 f(x) v = \lim_{t \downarrow 0} \frac{\big(\nabla f(x + tv) - \nabla f(x)\big)^T v}{t} \geq 0.$$

Since $v \in \mathbb{R}^d$ is arbitrary, $\nabla^2 f(x) \succeq 0$. This holds for every $x \in C$.

($\Longleftarrow$) Assume that $\nabla^2 f(z) \succeq 0$ for every $z \in C$. Fix $x, y \in C$, set $d = y - x$, and consider the restriction of f to the segment from x to y:

$$\phi(t) = f(x + td), \qquad 0 \leq t \leq 1.$$

Because C is convex, $x + td \in C$. By the chain rule,

$$\phi'(t) = \nabla f(x + td)^T d, \qquad \phi''(t) = d^T \nabla^2 f(x + td) d \geq 0.$$

By Taylor's theorem in one dimension there is a number $\tau \in (0, 1)$ with

$$\phi(1) = \phi(0) + \phi'(0) + \tfrac{1}{2}\phi''(\tau) \geq \phi(0) + \phi'(0).$$

Since $\phi(1) = f(y)$, $\phi(0) = f(x)$, and $\phi'(0) = \nabla f(x)^T (y - x)$, we obtain

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) \qquad \text{for all } x, y \in C. \tag{4}$$

Hence f is convex on C.

1. **Conceptual checks**

Determine whether each statement is *True* or *False*. Give a short justification; for a false statement, give a counterexample.

(a) If $f : \mathbb{R}^d \to \mathbb{R}$ is strictly convex and differentiable and $\nabla f(x^*) = 0$, then x^* is the unique global minimizer.

(b) If $f : \mathbb{R} \to \mathbb{R}$ is twice continuously differentiable, $f'(x^*) = 0$, and $f''(x^*) \geq 0$, then x^* must be a local minimizer.

(c) For a μ-strongly convex and L-smooth function, gradient descent with step size $1/L$ has rate factor $1 - \mu/L = 1 - 1/\kappa$, where $\kappa = L/\mu$. Thus a larger condition number gives a slower worst-case rate.

Solution: (a) True. Since every strictly convex function is convex, the first-order characterization of convexity gives, for every $y \in \mathbb{R}^d$,

$$f(y) \geq f(x^*) + \nabla f(x^*)^T (y - x^*) = f(x^*).$$

Hence x^* is a global minimizer.

It remains to prove uniqueness. Suppose, for contradiction, that there is another global minimizer $y^* \neq x^*$. Then

$$f(y^*) = f(x^*).$$

Since f is strictly convex, for every $\theta \in (0, 1)$,

$$f\big(\theta x^* + (1-\theta) y^*\big) < \theta f(x^*) + (1-\theta) f(y^*) = f(x^*).$$

This contradicts the fact that x^* is a global minimizer. Therefore no such y^* exists, and x^* is the unique global minimizer.

(b) False. Take $f(x) = x^3$ and $x^* = 0$. Then

$$f'(0) = 0, \qquad f''(0) = 0 \geq 0,$$

but 0 is not a local minimizer: for every sufficiently small $x < 0$,

$$f(x) = x^3 < 0 = f(0).$$

So the two conditions do not force a local minimum. A strict condition such as $f''(x^*) > 0$ is sufficient, but $f''(x^*) \geq 0$ alone is not.

(c) True. For gradient descent with step size $1/L$, strong convexity gives

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^k \big(f(x^{(0)}) - f(x^*)\big).$$

Since $\kappa = L/\mu$, the contraction factor is $1 - 1/\kappa$. A larger κ makes this factor closer to 1, so the worst-case rate is slower.

2. **Logistic regression: strict versus strong convexity**

Let

$$a_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y_1 = 1, \qquad a_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad y_2 = -1,$$

and

$$\ell(x) = \frac{1}{2} \sum_{i=1}^{2} \log\bigl(1 + e^{-y_i a_i^T x}\bigr), \qquad x \in \mathbb{R}^2.$$

Set $m_i(x) = y_i a_i^T x$ and

$$w_i(x) = \frac{e^{m_i(x)}}{(1 + e^{m_i(x)})^2}.$$

You may use

$$\nabla^2 \ell(x) = \frac{1}{2} \sum_{i=1}^{2} w_i(x) a_i a_i^T.$$

(a) Show that $\nabla^2 \ell(x) \succ 0$ for every finite x, and deduce that ℓ is strictly convex.

(b) For $x_t = t(1, -2)^T$, show that ℓ is not strongly convex and that $\inf \ell = 0$ is not attained. Explain why this does not contradict strict convexity.

Solution: (a) For every finite x we have $w_i(x) > 0$, because the numerator and the denominator are positive. For $z = (z_1, z_2)^T$,

$$z^T \nabla^2 \ell(x) z = \frac{1}{2} \Bigl[w_1(x) z_1^2 + w_2(x)(z_1 + z_2)^2 \Bigr].$$

Every term on the right is nonnegative, so $z^T \nabla^2 \ell(x) z \geq 0$. If this quantity is zero, then positivity of w_1, w_2 forces $z_1 = 0$ and $z_1 + z_2 = 0$, hence $z = 0$. Therefore

$$z^T \nabla^2 \ell(x) z > 0 \qquad \text{for every } z \neq 0,$$

that is, $\nabla^2 \ell(x) \succ 0$.

To pass from positive definite Hessians to strict convexity, take $u \neq v$ and set $g(\theta) = \ell\bigl(u + \theta(v - u)\bigr)$ for $\theta \in [0, 1]$. By the chain rule,

$$g''(\theta) = (v - u)^T \nabla^2 \ell\bigl(u + \theta(v - u)\bigr)(v - u) > 0,$$

because $v - u \neq 0$ and the Hessian is positive definite. Hence g is strictly convex on $[0, 1]$, and for $0 < \theta < 1$,

$$\ell\bigl(\theta u + (1 - \theta)v\bigr) = g(\theta) < \theta g(1) + (1 - \theta) g(0) = \theta \ell(u) + (1 - \theta)\ell(v).$$

So ℓ is strictly convex on $\mathbb{R}^2$.

(b) Along $x_t = (t, -2t)^T$ both signed margins are equal:

$$m_1(x_t) = y_1 a_1^T x_t = t, \qquad m_2(x_t) = y_2 a_2^T x_t = t.$$

Hence

$$w_1(x_t) = w_2(x_t) = \frac{e^t}{(1+e^t)^2} \longrightarrow 0 \qquad (t \to \infty).$$

With $e_2 = (0,1)^T = 0 \cdot a_1$ in the second coordinate, only the second sample contributes, so

$$e_2^T \nabla^2 \ell(x_t) e_2 = \tfrac{1}{2} w_2(x_t) \longrightarrow 0.$$

If ℓ were μ-strongly convex for some $\mu > 0$, the Hessian characterization would give $\nabla^2 \ell(x) \succeq \mu I$ for every x, so in particular $e_2^T \nabla^2 \ell(x_t) e_2 \geq \mu$ for every t. This contradicts the limit above. Therefore ℓ is not strongly convex.

Moreover,

$$\ell(x_t) = \tfrac{1}{2}\big[\log(1+e^{-t}) + \log(1+e^{-t})\big] = \log(1+e^{-t}) \longrightarrow 0.$$

Every term $\log(1 + e^{-m_i(x)})$ is strictly positive at every finite x, so $\ell(x) > 0$ for all finite x. Hence $\inf \ell = 0$, but the value 0 is never reached: $\arg\min \ell = \varnothing$. There is no contradiction with strict convexity. Strict convexity guarantees that a minimizer is unique *if it exists*; it does not guarantee existence on an unbounded domain.

3. **Conditioning and gradient descent**

For $\kappa \geq 1$, consider

$$f_\kappa(x) = \frac{1}{2}\big(x_1^2 + \kappa x_2^2\big), \qquad x = (x_1, x_2)^T \in \mathbb{R}^2.$$

(a) Determine the sharp strong-convexity constant μ, the sharp smoothness constant L, and the condition number.

(b) Run gradient descent with step size $\alpha = 1/L$. Derive the coordinate-wise update and hence an explicit formula for $x^{(k)}$ in terms of $x^{(0)}$.

(c) Explain, from the coordinate formula, why a large condition number slows gradient descent even though the high-curvature coordinate is reduced very quickly. Relate your explanation to the rate factor $1 - \mu/L$.

Solution: **(a)** The gradient and Hessian are

$$\nabla f_\kappa(x) = \begin{pmatrix} x_1 \\ \kappa x_2 \end{pmatrix}, \qquad \nabla^2 f_\kappa(x) = \begin{pmatrix} 1 & 0 \\ 0 & \kappa \end{pmatrix}.$$

The eigenvalues of the Hessian are 1 and κ, so the sharp constants are

$$\mu = 1, \qquad L = \kappa, \qquad \frac{L}{\mu} = \kappa.$$

(b) With $\alpha = 1/L = 1/\kappa$ the two coordinates decouple. For the first coordinate,

$$x_1^{(k+1)} = x_1^{(k)} - \frac{1}{\kappa}\, x_1^{(k)} = \left(1 - \frac{1}{\kappa}\right) x_1^{(k)}.$$

For the second coordinate,

$$x_2^{(k+1)} = x_2^{(k)} - \frac{1}{\kappa}\, \kappa x_2^{(k)} = x_2^{(k)} - x_2^{(k)} = 0.$$

So x_2 is removed in one step, and for every $k \geq 1$

$$x^{(k)} = \begin{pmatrix} \big(1 - \frac{1}{\kappa}\big)^k x_1^{(0)} \\ 0 \end{pmatrix}.$$

(c) The step size $\alpha = 1/L = 1/\kappa$ is exactly tuned to the curvature $L = \kappa$ of the second coordinate, so that coordinate is eliminated immediately. The same step size is much too small for the flat direction x_1, whose curvature is only $\mu = 1$. There the contraction factor per step is

$$1 - \frac{1}{\kappa} = 1 - \frac{\mu}{L},$$

which is close to 1 when κ is large. The flat direction therefore moves very slowly, and gradient descent needs many iterations. This is the geometric meaning of a large condition number.